3.3 Latent Knowledge Analysis

In this section, we provide more evidence of our hypothesis that FLU unlearning targets the source of intermediate latent knowledge. We analyze the Sports-Athlete-Editing task again here because the ground truth and the edited answers vary between one of only three possibilities.

We train logistic regression models (probes) (Alain & Bengio, 2018) on prompt activations following every model layer to predict the correct ground truth sport on the maintained set of athletes. This is possible because there are only three possible sports, so we can train binary classification probes for each sport and take the maximum classification over the sports. This is meant to discover internal representations of the true sport the model believes the answer to be: then, we apply these probes on the activations of the forget set of athletes.

We present graphs averaged over models and over 16 and 64 facts. Because layer is on the x-axis and models have different numbers of layers, averaging performance at a layer over multiple models isn't perfectly valid, so we also present probing graphs for each individual model and all localizations in Appendix A.7.4.

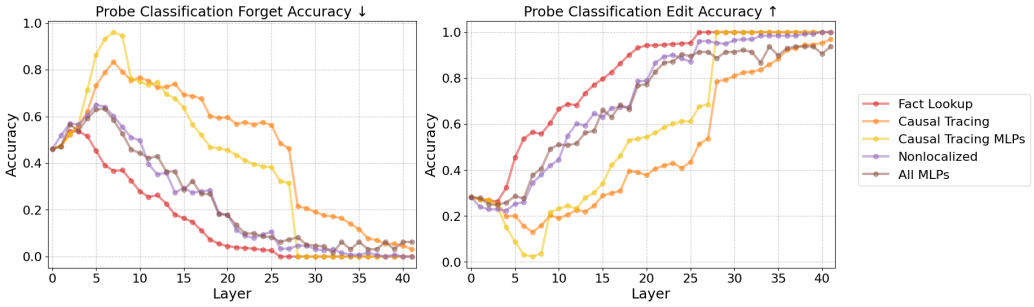

Figure 7: Linear probes applied to the forget set, classifying model activations after various layers. **(Left)** The line graph shows that some localizations still represent almost completely correct forget set knowledge in early layers, especially OT, while FLU localizations represent this original knowledge the least. **(Right)** The line graph shows that FLU localizations represent the edited rather than original answer earlier and more consistently throughout layers than any other localization.

Sports Results In Figure 7, the probes on FLU consistently predict the forget answer less and the edit answer more than in any other localization, especially in early layers. Furthermore, the FLU probe classifications for the most part monotonically converge from their nonzero starting accuracy to 0 (for forget accuracy) and 1 (for edit accuracy).

Every other localization has much higher peak probe classification forget accuracy in the early layers, especially the OT localizations which have peak classification forget accuracy of almost 100%. This strongly suggests that these models still significantly represent the ground truth answer rather than the edit answer in early layers.

3.4 The Role of Parameter Count

In this section, we perform weight-masking to quantify the size of edits with different localizations, as well as to investigate which factual mechanisms are targeted when editing with different localizations. We employ a weight masking technique involving training a binary differentiable mask over individual weights of the model within the localized components, inspired by weight pruning/masking work (Bayazit et al., 2023; Panigrahi et al., 2023). In this case, no weight updates are being performed. Rather, the mask turns a subset of the weights to zero.

Controlling for Parameter Count Although we already standardize the number of trainable parameters in most localizations, we additionally investigate if FLU editing is better than other localization techniques when controlling for the exact number of parameters that are masked.

We perform weight masking on the Sports-Unlearning, Sports-Athlete-Editing, and CounterFact-Editing tasks. Detailed results are reported in Appendix A.4. We find that when controlling for the size of the localization FLU is consistently more robust when subject to our suite of evaluations.

Other Localizations Affect the Extraction Mechanism After training weight masks, we analyze the proportion of each mechanism (fact lookup, attribution extraction) that is masked by each localization's weight mask in Appendix A.5. We demonstrate that OT methods and nonlocalized editing all modify a higher proportion of the extraction head/MLP parameters than the fact lookup mechanism parameters, supporting our claim that OT methods target extraction mechanisms rather than the fact lookup mechanisms needed for robustness.

4 Discussion

Recent work by Hase et al. (2023) argued that localization is not useful for model editing. Our findings demonstrate that the relationship between localization and fact editing/unlearning is more nuanced, and reveals that not all localization techniques are equal.

Our work evaluates the efficacy of different localization methods for modifying factual associations. We demonstrate clear benefits of localization for editing robustness through localized fine-tuning combined with FLU interpretability.

In Sections 3.3 and 3.4 and Appendix A.5, we provide evidence that OT and baseline approaches fail to be robust because they target easily-localizable and high direct logit importance extraction components that transform existing latent factual knowledge to the desired output format. This can fail to generalize to different input and output formats and does not target the source of knowledge in the model: other input/output formats can allow alternative mechanisms to extract this knowledge, and relearning can repair the original mechanisms to recover accuracy. In contrast, FLU mechanistic understanding allows us to target editing at the sites where knowledge is sourced, which robustly prevents that information from entering the latent stream in any format.

Our work also suggests unlearning/editing as a potential testbed for different interpretability methods, which might sidestep the inherent lack of ground truth in interpretability (Templeton et al., 2024). We hope our work provides a framework for evaluating localizations and explanations.

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